Weighted boundary costs and accepted local repairs

Research note

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Abstract

We separate two requirements for weighted rough-region estimates: control of a weight's change during an edit and control of its bias on frozen boundary labels. The former does not imply the latter, even for positive, local, gauge-invariant weights: an exterior-only tilt can make a rare frustrated boundary likely while remaining unchanged by the edit. We prove a one-region comparison under an explicit local marginal-density condition, and verify a different sufficient structural class without assuming that density condition: bounded reflection-symmetric interactions on elementary four-cells. In this class, weighted boundary-energy moments and local edit costs have constants proportional to the affected region, not the full volume. An accepted low-minimum repair has an output submeasure dominated by a small multiple of the same weighted measure; this permits composition of specified successful repairs. Exceptional retained labels do not admit that multiplication rule. All conclusions are for positive finite-lattice base measures. We do not establish a mandatory covering expansion, RG closure, cutoff comparison, or a continuum Yang–Mills theory. No novelty claim is made. The third-pass input has been adversarially reread and its geometric application made explicit; this is not independent expert or agent certification.

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1 Definitions and the third-pass input

Use $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, $L = 2^m$, $m \geq 2$, with $V = L^4$ sites, $4V$ links, and $6V$ plaquettes. Orient each link positively and assign $U_e \in \mathrm{SU}(2)$, with the inverse on its reverse. The reference measure is

product probability Haar. For $H \geq 1$ put

$$s(g) = 1 - \tfrac{1}{2}\operatorname{Tr} g = \tfrac{1}{2}\,\|\mathbf{1} - g\|_{\mathrm{op}}^2\,, \quad E = \sum_q s(U_q), \quad \mu_H(\,\mathrm{d}U) = Z_H^{-1} e^{-HE(U)}\,\mathrm{d}U.$$

For a fixed nonempty editable link set J, let $n = |J|$, let $P(J)$ contain every plaquette touching J, and set

$$E_J = \sum_{q \in P(J)} s(U_q), \qquad m_J(z) = \min_{x \in \mathrm{SU}(2)^J} E_J(x, z), \quad z = U_{J^c}.$$

Let F_J be the frozen links appearing on these plaquettes and $\mathcal{F}_J = \sigma(U_e : e \in F_J)$. Then m_J is continuous and $\mathcal{F}_J$-measurable, with $|F_J| \leq 18n$. This local sigma algebra is smaller than that of the complete frozen exterior.

Let $N_p(J)$ count plaquettes in $P(J)$ with operator curvature greater than $p/\sqrt{H}$, where $p > 0$. Define

$$\ell_H = \log 24 + \tfrac{3}{2}\log H, \quad a_H = \tfrac{3}{2} + \log 24 + \tfrac{3}{2}\log(8H), \quad b_H = \tfrac{1}{2}a_H + \frac{\log 2}{64}. \tag{1}$$

The third-pass note [1] proves

$$\mathbb{E}_{\mu_H} e^{HE_J/128} \leq e^{b_H n}, \qquad \mu_H(m_J > v) \leq \min\{1, e^{-Hv/128 + b_H n}\}. \tag{2}$$

It also constructs a Borel gauge-covariant kernel K_J which freezes J^c and samples probability Haar restricted to

$$B_J(z) = \{y : E_J(y, z) \leq m_J(z) + 12n/H\}.$$

Its edited-link density is at most $M_J = e^{\ell_H n}$. The minimizer-cap argument ensures the target has positive Haar volume at least M_J^{-1}; no measurable choice of a minimizer is needed. This is an existence construction, not an efficient algorithm for sampling or optimization.

For a positive integer k write

$$\begin{aligned} R_{J,k} &= \{N_p(J) \geq k\}, \qquad u = kp^2/(4H), \\ A_{J,k} &= R_{J,k} \cap \{m_J \leq u\}, \qquad B_{J,k} = \{m_J > u\}. \end{aligned} \tag{3}$$

Thus $R_{J,k} \subset A_{J,k} \cup B_{J,k}$. If $kp^2 \geq 96n$, every output from $A_{J,k}$ obeys

$$N_p(J; y) \leq 3k/4, \qquad H(E(x) - E(y)) \geq kp^2/8. \tag{4}$$

This reduces the total affected rough count, not necessarily each old rough plaquette; individual newly rough plaquettes are allowed.

2 Adversarial audit of the chessboard application

The analytic input behind (2) is the standard event chessboard inequality [2, §2.2, Theorem 2.4]. The closed-block convention and the fact that only the designated block-boundary reflections are needed are also explicit in [3, §5.3, Theorem 5.8]. We retain the power-of-two restriction, and use no assertion of independent blocks.

Here is a precise geometric encoding for the link-variable application. Put the variable for $(x, x + e_\mu)$ at $2x + e_\mu$ on the refined torus $(\mathbb{Z}/(2L)\mathbb{Z})^4$. These are exactly the points with one odd coordinate and the other three even. Fill the other refined sites with deterministic dummy variables. An original reflection $x_i \mapsto 2a - x_i$ becomes $v_i \mapsto 4a - v_i$. It preserves this parity pattern, swaps the two closed-half edge algebras, and inverts the group variable if its odd coordinate is i. Tangential

fixed-plane variables are pointwise fixed. Reflections in different coordinate directions commute, including their orientation inversions. No commutation of parallel reflections at different positions is asserted.

For the Wilson action no plaquette meets both strict halves. With z the fixed-plane links and x_+, x_- the remaining variables, the energy splits as $E_P(z) + E_+(x_+, z) + E_-(x_-, z)$ with the two half-energies reflections of each other. Thus for a bounded real positive-half function f, its reflection ϑf satisfies

$$\mathbb{E}_{\mu_H}[f\vartheta f] = Z_H^{-1} \int e^{-HE_P(z)} \left(\int e^{-HE_+(x_+,z)} f(x_+, z)\, \mathrm{d}x_+ \right)^2 \mathrm{d}z \geq 0. \tag{5}$$

Original side-two blocks become closed side-four refined blocks; the factor torus still has $M = V/16$ blocks. Their shared boundary variables are precisely the variables kept as z, not independent replicas. This checks the support and reflection-algebra assumptions used in the repeated Cauchy–Schwarz proof of the event chessboard estimate.

A prototype link has local starting coordinate 0 or 1 along its direction and 1 in the other three coordinates. All six star plaquettes have relative interiors inside its side-two block. Reflected stars in different blocks therefore have disjoint plaquette sets. The 64 classes are direction plus four starting-site parities. For each class the grid can be shifted so its links occupy distinct blocks; an odd longitudinal block changes the pulled-back prototype from 0 to 1. Both prototypes satisfy the same disjointness bound. This verifies the hypotheses for unequal thresholds at different stars.

Finally, a product cap of radius $(8H)^{-1/2}$ gives $Z_H \geq e^{-4Va_H}$. A disseminated event requiring star energy at least v forces total energy at least Mv, hence has chessboard weight at most $\min\{1, e^{-Hv+64a_H}\}$. Within a class, joint tails factorize *as an inequality*. Layer-cake integration and Hölder over the 64 classes give (2). The four occurrences of each plaquette in the uncolored sum of all stars are not discarded as if disjoint.

No mathematical error requiring a change to the third-pass proof was found in this bounded reread. Its restricted volume class, raw-marginal scope, overlap losses, and distinction between output-weighted integrals and probabilities remain necessary. This is a same-agent audit, not an independent certification. The accompanying exact finite-model and geometric tests are additional diagnostics, not proofs of the analytic chessboard theorem.

3 The missing marginal condition and a counterexample

Let $Q > 0$ be finite almost everywhere and satisfy $0 < \mathbb{E}_{\mu_H} Q < \infty$. Define

$$\nu_Q(\,\mathrm{d}U) = \frac{Q(U)}{\mathbb{E}_{\mu_H} Q} \mu_H(\,\mathrm{d}U), \qquad r_J = \frac{\mathbb{E}_{\mu_H}[Q \mid \mathcal{F}_J]}{\mathbb{E}_{\mu_H} Q}.$$

Then r_J is exactly the Radon–Nikodym density of the weighted F_J marginal relative to the raw one. For any $\mathcal{F}_J$-measurable event B,

$$\nu_Q(B) = \mathbb{E}_{\mu_H}[r_J \mathbf{1}_B]. \tag{6}$$

The normalization is the *full* expectation of Q. A conditional or restricted normalizer cannot be substituted without changing this law.

Proposition 3.1 (Edit stability alone gives no exceptional-label cost)**.** *There are strictly positive local gauge-invariant weights, constant on every J-edit, which assign any prescribed probability $s \in (0,1)$ to a given high-minimum boundary event of raw probability $t \in (0,1)$.*

Proof. For one link take $B = \{m_J > v\}$ with $0 < v < 6$. The balanced frustrated star has $m_J = 6$, identity staple data have $m_J = 0$, and both have open neighborhoods on the corresponding side of the threshold. The finite Wilson law has everywhere positive continuous density, so $0 < t = \mu_H(B) < 1$. The event depends only on the 18 frozen staple links and is gauge invariant. Set

$$Q = q\mathbf{1}_B + \mathbf{1}_{B^c}, \qquad q = \frac{s(1-t)}{(1-s)t}.$$

This weight is positive, finite and unchanged by any edited central-link value, but $\nu_Q(B) = qt/(qt + 1 - t) = s$. Its normalized local marginal density is s/t on B and $(1-s)/(1-t)$ on B^c. Hence, for $r > 1$,

$$\|r_J\|^r_{L^r(\mu_H)} = s^r t^{1-r} + (1-s)^r (1-t)^{1-r}. \tag{7}$$

The missing marginal cost becomes large as t decreases at fixed s, although the edit ratio stays exactly one. □

The phenomenon is not an artifact of discontinuous weights. Choose a continuous function $f(m)$ that vanishes for $m \le v$, equals one on $m \ge v' > v$ with $v' < 6$, and lies in $[0, 1]$. Then $Q_T = e^{Tf(m_J)}$ is continuous, positive, gauge invariant and edit-invariant, while $\nu_{Q_T}(B) \to 1$ as $T \to \infty$: a positive-measure subset of B has weight e^T, while B^c has weight one. No uniform amplitude bound in T or H is claimed.

Lemma 3.2 (Local marginal transfer)**.** *For $1 < r \le \infty$, let $\alpha = 1 - 1/r$ (with $1/\infty = 0$). If $\|r_J\|_{L^r(\mu_H)} \le e^{cn}$, then*

$$\nu_Q(B_{J,k}) \le \min\{1, e^{cn}\mu_H(B_{J,k})^\alpha\} \le \min\{1, \exp[-\alpha k p^2/512 + (c + \alpha b_H)n]\}. \tag{8}$$

Proof. Apply Hölder to (6), and then (2) at $v = kp^2/(4H)$. Using the bound without its truncation by one is valid; the final probability can again be truncated by one. □

This is an explicit sufficient hypothesis, not a proof that an arbitrary RG-generated weight satisfies it. It concerns a local projection of Q, not its whole-volume L^r norm. Equation (7) explains why some quantitative control of boundary bias is indispensable.

A directly verified elementary class is $Q = e^W$ with $\sup W - \inf W \le cn$: then $r_J \le e^{cn}$ and every edit ratio is at most e^{cn}. For example a perturbation supported on at most sn links, with total log-oscillation at most $c_0 sn$, has this property. *Local support alone* is not the bound, and adding an arbitrary exterior factor destroys the argument. An extensive bounded interaction usually has total oscillation of order V; the next structural class avoids importing that whole-volume loss into a local estimate.

4 An output domination lemma for positive weights

For a source event A, write

$$T_{J,A}(x, \mathrm{d}y) = \mathbf{1}_A(x) K_J(x, \mathrm{d}y).$$

This is an accepted-transition subprobability kernel, not the Markov kernel obtained by staying put on rejected sources.

Lemma 4.1 (Weighted accepted output)**.** *Suppose on source/kernel-output pairs from A,*

$$H(E(x) - E(y)) \ge g, \qquad Q(x) \le e^{dn} Q(y).$$

Then as measures,

$$\nu_Q T_{J,A} \le C_J \nu_Q, \qquad C_J = \exp[-g + (\ell_H + d)n]. \tag{9}$$

In particular $\nu_Q(A) \le \min\{1, C_J\}$*, and for every measurable* f *with* $\mathbb{E}_{\nu_Q}|f| < \infty$*,*

$$\frac{1}{Z_Q}\int_A e^{-HE(x)} Q(x)(K_J|f|)(x)\,\mathrm{d}x \le C_J \mathbb{E}_{\nu_Q}|f|, \quad Z_Q = \int e^{-HE} Q. \tag{10}$$

Proof. Write $x = (\xi, z)$, $y = (\eta, z)$ and let $k_z(\xi, \eta) \le M_J$ be the edited-output density. On the indicated pairs, $e^{-HE(\xi,z)}Q(\xi, z) \le e^{-g+dn}e^{-HE(\eta,z)}Q(\eta, z)$. For a nonnegative output test function f, insert this inequality, bound k_z by M_J, and integrate the original ξ over probability Haar. Its mass is at most one. Tonelli leaves at most $e^{-g+dn}M_J \int f\,\mathrm{d}\nu_Q$, proving measure domination. Since K_J is a probability kernel, the total mass on the left is $\nu_Q(A)$. Apply the result to $|f|$ for (10). □

The positive base weight remains at the actual output after the comparison. Signed or complex *test functions* may be estimated by their absolute values; signed/complex base densities are not allowed. For contrast the earlier literal integral $Z_Q^{-1}\int_A e^{-HE(x)}(K_J Q)(x)\,\mathrm{d}x$ has a bound without e^{dn}, but is not in general $\nu_Q(A)$. These expressions are not identified.

Theorem 4.2 (One-region comparison with declared weight hypotheses)**.** *Assume* $kp^2 \ge 96n$*, the edit-ratio bound of Lemma 4.1 on* $A_{J,k}$*, and the local marginal bound of Lemma 3.2. Then*

$$\nu_Q(R_{J,k}) \le \min\left\{1, e^{-kp^2/8+(\ell_H+d)n} + e^{-\alpha kp^2/512+(c+\alpha b_H)n}\right\}. \tag{11}$$

The low-minimum transition has the output domination (9) *with* $g = kp^2/8$ *and reduces its rough count as in* (4)*.*

Proof. Use $R_{J,k} \subset A_{J,k} \cup B_{J,k}$ and apply the two lemmas in the *same* measure ν_Q. They control different branches, with separate hypotheses. □

5 A verified class of extensive positive perturbations

For each $x \in \Lambda_L$ let $C_x = x + [0, 1]^4$ be an elementary four-cell, including its 32 links. There are V such cells. Each link belongs to exactly eight cells. We admit

$$Q = e^W, \qquad W = \sum_C \phi_C, \tag{12}$$

where ϕ_C is a bounded real Borel function of the links in C, gauge invariant, and $\operatorname{osc}\phi_C \le \varepsilon$. Require translation covariance and covariance under every integer-site coordinate reflection, including orientation reversal of links: $\phi_{\vartheta C}(\vartheta U) = \phi_C(U)$. No smoothness or sign restriction on the real ϕ_C is needed. The Boltzmann weight itself is always positive. The amplitude $\varepsilon \ge 0$ may depend on H; useful small-field budgets will require it to be sufficiently small relative to p^2, not necessarily an absolute small constant.

Subtract $\inf\phi_C$ from each term. These are configuration-independent constants and do not change ν_Q or edit ratios; covariance is retained. Thus we may assume

$$0 \le \phi_C \le \varepsilon, \qquad 0 \le W \le \varepsilon V. \tag{13}$$

Although the global range is extensive, the resulting local estimates below are not multiplied by $e^{\varepsilon V}$.

Lemma 5.1 (Weighted reflection and local edit cost)**.** *The measure* $\nu_W = Z_W^{-1} e^{-HE+W}\,\mathrm{d}U$ *is site-reflection positive. For any pair of configurations differing only on* J*,*

$$|W(x) - W(y)| \le 8\varepsilon n. \tag{14}$$

Proof. An elementary cell has extent one in each coordinate, so its link support cannot meet both strict halves separated by an integer-coordinate plane. All terms in W therefore split into plane and reflected half terms, just as the Wilson plaquettes do. The positive fixed-plane factor and the square in (5) are replaced by

$$e^{-HE_P(z)+W_P(z)}, \qquad \left(\int e^{-HE_+(x_+,z)+W_+(x_+,z)} f(x_+,z)\,\mathrm{d}x_+\right)^2.$$

This proves positivity. Only cells containing a link of J can change under the edit, and there are at most $8n$ such cells. Sum their oscillation bounds. No independence of cells is used. □

Theorem 5.2 (Weighted local energy moments)**.** *For the class* (12)*, for every fixed* J *and* $0 < \theta < 1$*,*

$$\mathbb{E}_{\nu_W} e^{\theta H E_J/64} \le \exp\left[n\left(\theta(a_H + \varepsilon/4) - \frac{\log(1-\theta)}{64}\right)\right]. \tag{15}$$

The same bound holds with E_J *replaced by* m_J*. Set* $\widehat{b}_H = b_H + \varepsilon/8$*. Then*

$$\nu_W(E_J \ge v),\ \nu_W(m_J > v) \ \le\ \min\{1, e^{-Hv/128 + \widehat{b}_H n}\}. \tag{16}$$

These constants are uniform in the dyadic volume and all admitted potentials with the given oscillation bound.

Proof. The centered weight satisfies $W \ge 0$, so the product cap still gives $Z_W \ge e^{-4Va_H}$. A disseminated star event forcing $E \ge Mu$ has numerator at most $e^{-HMu+\varepsilon V}$. Its chessboard weight is therefore at most

$$q_{H,\varepsilon}(u) = \min\{1, \exp[-Hu + 64a_H + 16\varepsilon]\}. \tag{17}$$

Lemma 5.1 licenses the same event chessboard inequality in ν_W. The geometric disjointness and the 64 classes are unchanged.

For completeness, write $K = 64a_H + 16\varepsilon$ and $T_e = HE_e$. Within a class, joint tails are bounded by products of $q(t) = \min(1, e^{K-t})$. Expanding $e^{\theta T_e} = 1 + \int_0^\infty \theta e^{\theta t} \mathbf{1}_{T_e > t}\,\mathrm{d}t$ in a finite product and applying Tonelli gives

$$\mathbb{E}_{\nu_W} e^{\theta \sum_{e\in S} T_e} \le \left(1 + \int_0^\infty \theta e^{\theta t} q(t)\,\mathrm{d}t\right)^{|S|} = \left(\frac{e^{\theta K}}{1-\theta}\right)^{|S|}.$$

Since $E_J \le \sum_{e\in J} E_e$, Hölder with 64 equal exponents proves (15). Pointwise $m_J \le E_J$ and Markov's inequality at $\theta = 1/2$ prove (16). □

This theorem proves the needed exceptional-label tail *directly in the weighted measure*. It does not prove, and does not require, the local likelihood-ratio hypothesis of Lemma 3.2 relative to μ_H. These are two distinct sufficient routes.

Corollary 5.3 (Verified one-region weighted comparison)**.** *For (12), if $kp^2 \geq 96n$, then*

$$\nu_W(R_{J,k}) \leq \min\left\{1, e^{-kp^2/8+(\ell_H+8\varepsilon)n} + e^{-kp^2/512+\widehat{b}_H n}\right\}. \tag{18}$$

The low-minimum output submeasure is dominated by the first exponential times ν_W, for any nonnegative output test function. The exceptional label is measured in that same ν_W. If also

$$kp^2 \geq 16n(\ell_H + 8\varepsilon), \qquad kp^2 \geq 1024n\widehat{b}_H, \tag{19}$$

then the first term is at most $e^{-kp^2/16}$ and the second at most $e^{-kp^2/1024}$.

Proof. Combine (14) with Lemma 4.1 on $A_{J,k}$. For $B_{J,k}$ apply (16) with $v = kp^2/(4H)$. The numerical absorptions follow directly from (19). □

As a direct probability estimate the split is weaker than the consequence $\nu_W(R_{J,k}) \leq \min\{1, e^{-kp^2/256+\widehat{b}_H n}\}$ of (16). Its additional content is the *accepted output domination*, with a retained exceptional branch in the same measure.

A nontrivial explicit example. Take

$$W_\gamma = \gamma \sum_q s(U_q)^2, \qquad \gamma \in \mathbb{R}. \tag{20}$$

Each plaquette belongs to four elementary four-cells, and a cell contains 24 plaquettes. The decomposition $\phi_C = (\gamma/4)\sum_{q\subset C} s(U_q)^2$ has $\operatorname{osc}\phi_C \leq 24|\gamma|$, since $0 \leq s^2 \leq 4$. Thus $\widehat{b}_H = b_H + 3|\gamma|$ is valid. Directly summing only the affected plaquettes improves the edit constant to

$$|W_\gamma(x) - W_\gamma(y)| \leq 4|\gamma||P(J)| \leq 24|\gamma|n.$$

Use this in place of $8\varepsilon n$ in (18) and (19). Both signs of γ are allowed. This is not just a change of the coefficient of $s(U_q)$ in the continuous SU(2) model. It is an example of the proved class, not a claim that exact RG stays inside it.

6 Counting and what composition actually permits

The link-sharing-a-plaquette graph has degree at most 18. For the actual rough-link component through a fixed root e_0, the same fixed-subset counting argument as in [1] now gives

$$\nu_W(|C_p(e_0)| \geq n) \leq \min\{1, 18^{2(n-1)} e^{-n[p^2/1024-\widehat{b}_H]}\}. \tag{21}$$

Every selected link has a rough witness, and a plaquette witnesses at most four links; no boundary compatibility or independence is assumed. The total weight W is fixed before choosing supports. Rarity is not re-evaluated under a separately chosen measure for each region. For $\varepsilon_H = O(\log H)$ and $p = (\log H)^P$, $P > 1/2$, the component and one-region absorption budgets hold eventually when $k \geq n/4$. This is only a large-H arithmetic assertion in the declared model class.

Proposition 6.1 (Composition of accepted repairs)**.** *Fix one positive base measure ν_Q. Suppose nonnegative subprobability kernels T_i satisfy $\nu_Q T_i \leq c_i \nu_Q$. Then*

$$\nu_Q T_1 \cdots T_r \leq \left(\prod_{i=1}^r c_i\right) \nu_Q. \tag{22}$$

In particular this applies to the low-minimum kernels in Corollary 5.3 on predetermined supports, with their gates A_{J_i,k_i} evaluated on each actual intermediate state. *No disjointness of affected plaquettes is required for this statement.*

Proof. A nonnegative kernel preserves order of measures. Apply T_2 to $\nu_Q T_1 \le c_1 \nu_Q$, then use $\nu_Q T_2 \le c_2 \nu_Q$, and iterate. For the asserted application, every one-step energy drop is proved against its actual source by (4); gains are not all measured against the common original configuration. □

For adaptive support choice from a fixed finite or countable family, partition the source-selection events and use the corresponding accepted kernels. Their sum is dominated by $\sum_J c_J$ times the base measure. For connected J of size n through a fixed root with a uniform $k \ge n/4$ budget, there are at most $18^{2(n-1)}$ terms. This explicitly pays the support count. A choice over arbitrary roots costs its additional locations. A policy depending on history needs a valid bound for each encoded history or a genuine aggregate kernel bound; the fixed-family statement does not silently supply one.

Corollary 6.2 (A counted adaptive accepted kernel)**.** *Fix a root e_0 and the cell-weight measure ν_W. On each source choose its current nonempty rough-link component $J = C_p(e_0)$ and set $k_J = \lceil |J|/4 \rceil$. Apply K_J only when $m_J \le k_J p^2/(4H)$, and otherwise give this accepted kernel T mass zero. If $p^2 \ge 384$ and*

$$d_H = p^2/32 - \ell_H - 8\varepsilon > 2\log 18,$$

then

$$\nu_W T \le c_H \nu_W, \qquad c_H = \frac{e^{-d_H}}{1 - 18^2 e^{-d_H}}. \tag{23}$$

Thus $\nu_W T^r \le c_H^r \nu_W$. In particular $c_H < 1$ if $d_H > \log 325$. This covers only paths on which all r selected repairs are accepted; the root and weight remain fixed.

Proof. The component has at least k_J rough witnesses and $k_J p^2 \ge 96|J|$, so every accepted fixed-J transition has domination constant at most $e^{-d_H|J|}$. The finite collection of component-selection events partitions the nonempty sources. Count its connected supports through e_0 by $18^{2(n-1)}$ and sum the geometric series, extending the finite size range to infinity for an upper bound. This is an aggregate one-step kernel estimate, not a conditional Gibbs calculation. Apply Proposition 6.1. The condition for $c_H < 1$ is $325 e^{-d_H} < 1$. □

This construction recalculates the component from the actual output at its next step. It may introduce individual new rough plaquettes and change component geometry; the aggregate estimate already counts those possibilities at each accepted step. It is not a mandatory-inventory preserving repair and does not treat the retained-exception branches.

Unconditional sublevel output. For the cell class, $E(y) \le E(x) + 12n/H$ holds for every source/output pair of the ungated K_J. Thus the same domination proof with $g = -12n$ gives

$$\nu_W K_J \le e^{(12+\ell_H+8\varepsilon)n} \nu_W. \tag{24}$$

This permits specified ungated edits with an *accumulating* density cost, not cost-free reuse of the original measure.

Retained exceptional labels do not multiply. The identity branch $S_B(x, \mathrm{d}y) = \mathbf{1}_B(x)\delta_x(\mathrm{d}y)$ obeys $\nu_Q S_B = \mathbf{1}_B \nu_Q$, which is not dominated by $\nu_Q(B)\nu_Q$ if $0 < \nu_Q(B) < 1$. Applying it twice still has mass $\nu_Q(B)$, not $\nu_Q(B)^2$. A small marginal probability is not the small *operator* domination needed in Proposition 6.1. Joint exceptional events need a joint-energy/overlap estimate or additional decorrelation; no product of one-label tails is asserted. Even disjoint link edits can share affected plaquettes, and a history can revisit the same boundary defect.

For fixed sets J_i, one available joint input remains

$$\mathbb{E}_{\nu_W} \exp\left[\frac{H}{128 m_*} \sum_i m_{J_i}\right] \le e^{\widehat{b}_H |\bigcup_i J_i|}, \qquad m_* = \max_q |\{i : q \in P(J_i)\}|,$$

by $\sum_i m_{J_i} \le m_* E_{\cup_i J_i}$ and Theorem 5.2. It concerns labels evaluated on *one* configuration, not a sequence of changed exteriors.

7 Limits and the next precise obligation

Why the hypotheses cannot be discarded. Proposition 3.1 disproves an exceptional-label bound from local edit stability alone. Reflection symmetry and positivity of a weight alone also do not imply reflection positivity: two symmetric spins $\sigma_\pm \in \{-1, 1\}$ with weight $e^{-\tau\sigma_+\sigma_-}$, $\tau > 0$, have $\mathbb{E}[\sigma_+ \vartheta \sigma_+] = -\tanh\tau < 0$. The interaction crosses both strict halves. This simple example is not a counterexample to the cell-local theorem; it explains why a longer interaction needs its own positivity proof.

The weight $Q = e^{HE}$ cancels Wilson suppression. It even has a cell representation with $\varepsilon = 12H$, by distributing plaquette energies over their four cells. Our resulting constants are then too large to imply small rough probabilities: $\widehat{b}_H \ge 3H/2$, while $p^2 < 4H$ in the nonempty rough regime. Thus the admitted-class theorem does not claim a useful bound at arbitrary perturbation amplitudes.

The frustrated star, padded isolated defect, flat winding holonomy, and the overlapping-edit example from the preceding notes remain valid. Neither rarity nor a reduction in rough count implies an all-small boundary-compatible filling or preservation of a prescribed inventory. The probability statements remain restricted to periodic dyadic cubes, not arbitrary frozen boundary conditions or all Gibbs states.

What is now discharged. For the explicitly admitted positive cell-interaction class, both branches of a one-region comparison are controlled in the *same* weighted measure with region-proportional costs. The accepted low-minimum output has enough measure domination to compose specified successful repairs. For general positive weights the exact missing local marginal-density hypothesis is isolated and shown not to follow from edit stability.

What remains. The spin-model endpoint comparison inspected in [4, §Integrated comparison of endpoint densities, equations labeled `eq:orig-16` and `eq:orig-22`] uses a mandatory covering sum with fixed inventories and signed/complex activities. Our cellwise exponential class is not that covering class and is not proved invariant under a gauge-theory RG step. Before claiming a cutoff comparison, one must establish a suitable positive endpoint class, prove its boundary marginal or energy-moment control with uniform constants, and treat the exact covering identities, residual inventories and all eligibility queries. Signed restricted sums must not be used as normalizers.

A concrete next deliverable is to test one explicitly specified coarse-graining step or covering-weight construction against these two requirements: a local edit cost and a uniform exceptional-label

estimate in its *positive complete endpoint measure*. If elementary-cell support or the declared reflection structure is lost, prove a replacement hypothesis instead of assuming closure. Composition involving retained exceptional labels and adaptive histories also requires a joint bound rather than repeated marginal tails. The fixed-physical-size defect test, nontrivial continuum fields and reconstruction remain downstream.

8 Checks and scope of verification

The accompanying standard-library script tests exact finite conditional marginals, normalized reweighting, local likelihood norms, input/output weight distinctions, domination and composition of accepted kernels, cell incidence and reflection geometry, and noncommuting quaternion-group link identities. A two-dimensional finite $\mathbb{Z}_2$ torus is enumerated exactly for weighted event chessboard diagnostics. These finite models do not calculate the four-dimensional SU(2) probabilities and do not prove universal analytic assertions. The earlier Python checks and standalone Lean scalar certificates are retained and rerun; no new analytic or gauge result is Lean-formalized. Independent review remains outstanding.

References

[1] *Marginal boundary energy costs in the Wilson model*, third-pass local research note, October 9, 2026, `../boundary_cost/boundary_cost.tex`. Reviewed source SHA-256 is recorded in `audit_third_pass.md`.

[2] M. Biskup and R. Kotecký, *Forbidden gap argument for phase transitions proved by means of chessboard estimates*, Commun. Math. Phys. **264** (2006), 631–656, arXiv:math-ph/0505011, author manuscript §2.2, Theorem 2.4, (2.10).

[3] M. Biskup, *Reflection positivity and phase transitions in lattice spin models*, author lecture notes, §5.3, Theorem 5.8, (5.35), pp. 43–46.

[4] OpenAI, *The canonical massive continuum limit of the two-dimensional $O(3)$ model*, October 4, 2026, supplied repository, `build/sections/comparison.tex`. Inspected comparison requirements; no transfer of its continuum result to the Wilson model is assumed.